Unsupervised Extractive Opinion Summarization based on Text Simplification and Sentiment Guidance

Anonymous ACL submission

Abstract

Unsupervised opinion summarization aims to extract representative content from a set of reviews without relying on golden references. Traditional unsupervised methods often struggle with non-consensus opinions and lack conciseness in the extracted summaries, which may prevent users from making swift and informed decisions. To tackle these challenges, we propose a novel two-stage unsupervised opinion summarization method based on text simplification and sentiment guidance. In the first stage, we leverage a pre-trained language model to simplify complex sentences into concise and clear forms. In the second stage, our method identifies and clusters sentences based on sentiment information. We then extract summary sentences that align with the overall sentiment tendency, ensuring consistency and representativeness. Experimental results on the Space and Amazon benchmark datasets demonstrate performance improvements, confirming the efficacy of our approach in addressing the identified challenges.

1 Introduction

Opinion summarization is a process of condensing multiple user comments into an expression of opinions that is both concise and consistent(Amplayo et al., 2021b; Amplayo and Lapata, 2020; Bražinskas et al., 2020a,b; Amplayo and Lapata, 2021; Angelidis and Lapata, 2018; Chu and Liu, 2019; Suhara et al., 2020; Amplayo et al., 2021a). Due to the absence of human-written summaries, a predominant approach in research utilizes unsupervised methods. Depending on their approach, opinion summarization technologies can be categorized into two types: abstractive summarization and extractive summarization.

In abstractive summarization, previous studies typically train supervised models on synthetic datasets(Amplayo et al., 2021b; Amplayo and Lapata, 2020; Bražinskas et al., 2020a; Amplayo et al., 2021a; Elsahar et al., 2021; Wang and Wan, 2021; Louis and Maynez, 2023; Ke et al., 2022; Bražinskas et al., 2022; Oved and Levy, 2021), with the success largely depending on the quality of these datasets. Some approaches leverage aggregated sentence representations from autoencoders to generate summaries(Chu and Liu, 2019; Isonuma et al., 2021). In contrast, extractive summarization involves two critical steps: (a) obtaining representations of comment sentences, and (b) employing algorithms to select the most representative set of sentences(Amplayo et al., 2021a; Angelidis et al., 2021; Chowdhury et al., 2022). Our research focuses specifically on enhancing the second step by refining the selection process to improve the quality of the extracted summaries.

Previous extractive methods often produce summaries containing non-consensus opinions, as shown in Figure 1. We hypothesize that this issue arises from the model's inability to accurately evaluate the importance of non-consensus opinion sentences and effectively distinguish between conflicting opinions. Observations indicate that conflicting sentences typically exhibit opposite sentiment information. Our method utilizes the sentiment information from comment sentences to enhance the selection of summary sentences. Experiments reveal that sentiment features are crucial for improving the quality of opinion summarization. Additionally, we observe that some original reviews contain complex sentences, which can lead to summaries that are neither concise nor succinct. Figure 3 exemplifies the process of simplifying complex review sentences. In the subsequent sections, we will demonstrate the effectiveness of our text simplification technique in the context of opinion summarization.

To address the aforementioned issues, we propose TSSG, an extractive summarization method based on Text Simplification and Sentiment Guidance. Specifically, we first perform text simplifi-

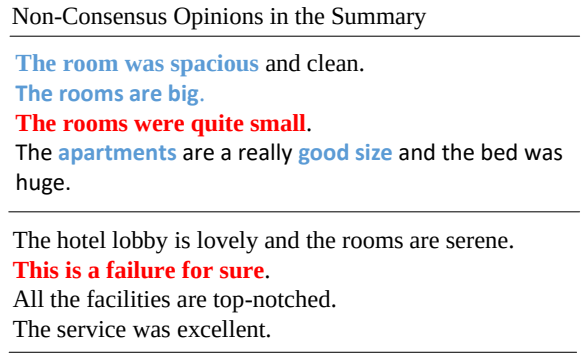


Figure 1: Examples of non-consensus Opinions in summary. In the first example, we use red and blue to label non-consensus opinions regarding a specific aspect, such as room size. In the second example, we use red to label non-consensus opinions.

cation on the set of reviews, converting complex texts into simpler forms. Next, We employ a sentiment analyzer to detect the sentiment polarity of the reviews and group them accordingly. Finally, we extract the most representative sentences from these groups based on the overall sentiment distribution to compose the opinion summaries. Our experiments demonstrate that TSSG outperforms competing baselines on the Space and Amazon datasets(Angelidis et al., 2021; Bražinskas et al., 2020b). Our contributions are as follows:

- We applied text simplification techniques to the field of opinion summarization and achieved good results.
- By leveraging the sentiment information of review sentences, we effectively eliminated non-consensus opinions from the summaries.
- We conducted extensive analytical experiments to validate the performance of TSSG.

2 Related work

2.1 Text Simplification

The goal of sentence simplification is to transform original (often complex) sentences into shorter, more understandable ones through various operations such as deletion, addition, and splitting of words and phrases(Sun et al., 2021).

Large pre-trained language models excel in tasks such as question answering, summarization, and simplification. Some studies have utilized models like T5 , along with controlled markers, to accomplish simplification tasks(Sheang and Saggion, 2021; Raffel et al., 2020). Inspired by these approaches, we integrated a simplification component into opinion summarization and achieved the best performance in our experiments.

2.2 Extractive opinion summarization

The general paradigm of extractive opinion summarization involves first learning representations of comment sentences and then extracting important sentences using various algorithm(Angelidis et al., 2021; Angelidis and Lapata, 2018; Chowdhury et al., 2022; Zhao and Chaturvedi, 2020; Li et al., 2023; Erkan and Radev, 2004). In the second step, methods such as the word-frequency-based PageRank algorithm are used to generate comment centroids and select comment segments(Erkan and Radev, 2004). Some approaches define relevance scores based on the distance between sentences and the average semantic value(Chowdhury et al., 2022), their proximity to aspect representations(Angelidis et al., 2021), or aspect specificity(Angelidis and Lapata, 2018; Zhao and Chaturvedi, 2020) to identify significant sentences. For instance, SemAE implicitly captures semantic information from comments using dictionary learning and selects sentences close to the mean as summary information(Chowdhury et al., 2022). HERCULES transforms sentence encodings into a hierarchical discrete latent space and extracts opinion summaries based on encoding frequency(Hosking et al., 2023). TokenCluster clusters sentences based on noun phrases and adjective roots and selects representative sentences from each cluster(Li et al., 2023). While previous extraction methods are efficient and easy to interpret, they often overlook the influence of sentiment information on opinion summarization.

3 Methodology

We propose a two-stage opinion summarization extraction framework, TSSG, as illustrated in Figure 2. TSSG consists of two main components: a text simplifier and a summary extractor. In the first stage, the text simplifier performs simplification operations on the text. In the second stage, the summary extractor conducts sentiment analysis on sentences and extracts significant sentences based on sentiment guidance to form the summary.

In this section, we first define the problem in 3.1. Next, we introduce the workflow of the text simplifier in 3.2. Following that, we classify the

sentiment polarity of comment sentences and select representative sentences to form a general summary in 3.3. Finally, we describe the process for generating aspect-specific summaries in 3.4.

3.1 Problem Definition

Given a corpus of reviews for a set of entities (e.g., products, hotels). Let e represent an entity. We use $\mathbf{S}_e = \{s_1, s_2, ...s_N\}$ to denote the set of sentences in all reviews for e,and R_e to cover a range of aspects related to certain domains. Our goal is to extract sentences from S_e to form two types of opinion summaries: (a) a general summary, which involves selecting a subset of sentences $O_e \in S_e$ that accurately represents the reviews in R_e ; (b) an aspect summary, where the generated summary $O_e^{(a)} \in S_e$ focuses on specific aspect $a \in A$.

3.2 Text Simplification Module

Recently, the application of large pre-trained language models to the text simplification field has shown outstanding performance(Sheang and Saggion, 2021; Blinova et al., 2023; Raffel et al., 2020). Inspired by this success, we utilize BART(Lewis et al., 2020) as the pre-trained model for sentence simplification in our text simplification module. We fine-tune BART using pre-processed datasets from document-to-document simplification tasks, specifically D-Wikipedia and Wiki-Doc(Sun et al., 2021; Kauchak, 2013).

We define the ideal length L of a simple sentence as 8 to 20 words. This definition is based on analyzing 330 manually written summaries (180 from Amazon and 150 from Space), where more than 85% of the sentences were shorter than 20 words. These suggests that human-written, concise opinion summary sentences typically do not exceed 20 words. Additionally, we expect the simplified sentences not to contain complex words. To achieve this goal, we fine-tune the generative model BART. Specifically, we designed a two-step process to obtain simple sentences.

For any sentence S_i in the sentence set S_e that exceeds 20 words, we first simplify it using BART.

$$S_i^{'} = LLM(S_i). \tag{1}$$

Here, LLM refers to the fine-tuned BART model, and $S_i^{'}$ represents the initially simplified sentence. Ideally, $S_i^{'}$ no longer contains complex words or phrases.

Secondly, we check the length of $S_i^{'}$ again. If it still exceeds 20 words, we split the sentence $S_i^{'}$ further. We use simple rules to split the sentence, based on conjunctions, periods, and commas, breaking $S_i^{'}$ into shorter segments from left to right. If any resulting segment is shorter than 3 words, it will be discarded, as we believe that extremely short sentences are unlikely to convey effective opinion information. An example of sentences processed by the simplification module is shown in Figure 3.

3.3 Sentence Extraction Module

3.3.1 sentiment Analysis

Intuitively, comments with similar sentiment orientations are less likely to conflict in content compared to comments with opposing sentiment orientations. Therefore, we customize sentiment analysis as a text classification task, categorizing the sentiment polarity of each review sentence. The sentiment label for the review set $S_{1:n}$ is calculated as follows:

$$P_i^{sen} = SLM\left(S_i\right) \tag{2}$$

$$Sen_i = \arg\max_j P_i^{sen}(j) \tag{3}$$

Here, S_i represents the i review. $SLM(\cdot)$ is a pre-trained model with a linear classification head. $P_i^{sen} \in R^K$ denotes the probability distribution of sentiment categories for the review sentence S_i. The category with the highest probability is assigned as the sentiment label Sen_i. We fine-tune the SLM using Space and Amazon data with sentiment labels. The loss is calculated using negative log-likelihood. We can obtain the sentiment labels for all review sentences through the fine-tuned BART model. Each review sentence will receive a sentiment polarity label. We set the number of sentiment categories to K. The review set S will then be divided into K subsets. Each subset will contain a portion of review sentences, denoted as S_j^e, under each sentiment category.

3.3.2 Sentence Ordering

We need to obtain the representation a_i for each sentence. In this paper, we employ the SemAE model(Chowdhury et al., 2022) to obtain the representation $a_i \in R^D$ for each sentence. After obtaining the representation for each review sentence, we need to compute the similarity of each sentence with the mean. We calculate the similarity of each

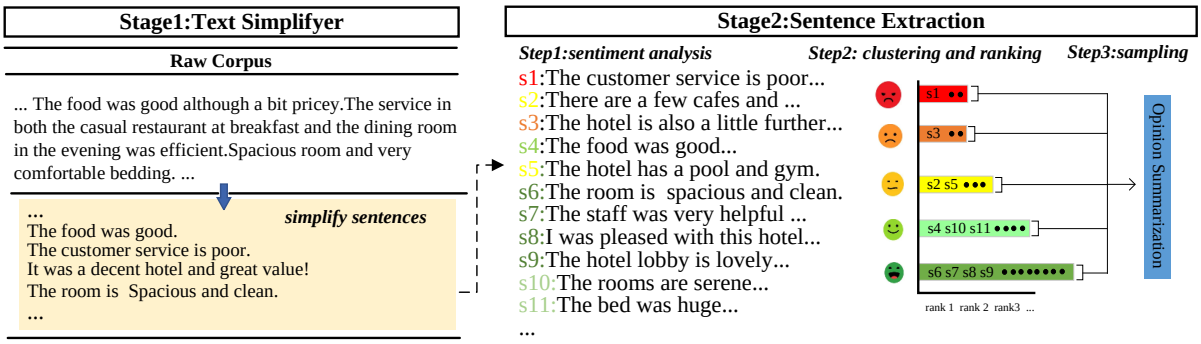


Figure 2: Text Simplification and Sentiment Guidance (TSSG) framwork.

sentence with the mean for each category and consider it as the relevance score $R(\cdot)$. Then, we sort the sentences in the sentiment set based on the relevance score. The calculation process of $R(\cdot)$ is as follows:

$$\mathrm{R}\left(\alpha_i, \bar{a}_j\right) = -\mathrm{KL}\left(\alpha_i, \bar{a}_j\right), \alpha_i \in a_j^{(e)} \quad (4)$$

Where $a_j^{(e)}$ represents the set of sentiment sentences of the j-th category. Where a_i represents the set of sentiment sentences of the i-th category. We use KL divergence to measure the similarity between the representations of two sentences.

To ensure that the selected sentences represent the overall consensus sentiment tendency of the entire review. We introduce a sentiment information penalty term to control the relevance score of non-consensus opinion sentences.

$$\mathrm{R}\left(\alpha_i, \bar{a}_j\right) = \mathrm{R}\left(\alpha_i, \bar{a}_j\right) - \beta \cdot KL\left(p_i, w\right) \quad (5)$$

$$P^{(e)} = \sum_{i=1}^{n} P_i \quad (6)$$

$$w_j = \frac{P_j^{(e)}}{\sum_{j=1}^{k} P_j^{(e)}} \quad (7)$$

Where $P^{(e)} \in R^K$ is the sum of sentiment distributions for all sentences. Where $w \in R^K$ is the result of normalizing $P^{(e)}$. We consider it as the overall sentiment tendency of the users' reviews towards the entity e. Where β is the weight of the penalty term. We still use KL divergence to measure the score between the sentence and the overall sentiment tendency w.

3.3.3 Sentence Extraction

Next, we will select a subset $S_{1:n}$ from the K subsets as the opinion summary. We define the following three sampling strategies:

1. **Random sampling :** From each set, a fixed number λ of sentences are randomly selected to form $S_{1:n}$.

2. **Equal Sampling :** From each set, we extract the top $N \times \varphi$ sentences based on their relevance scores to form $S_{1:n}$. The value of φ ranges from 0 to 1.

3. **Weighted sampling :** First, we calculate w, considering it as the overall sentiment towards entity e from user reviews, as shown in formula (7). Then, we set the total number of extracted sentences as n. Using w as the sampling proportion, the number of sentences to be extracted from the i set is $n \times w_i$. Finally, sentences are extracted from each set to obtain $S_{1:n}$.

After extracting the summary candidate sentence set using one of the above algorithms, the number of sentences in the set may still be large. Therefore, we calculate the importance score of candidate summary sentences with respect to the mean representation again. The calculation process is as shown in equation 5. Finally, we select the top N sentences with the highest $R(\cdot)$ scores as the summary.

3.4 Summary of Specific Aspects

To obtain aspect-based opinion summaries, we refer to previous research for guidance. We assume that sentences describing the same aspect will mention keywords belonging to the same aspect category. We use a set of aspect-related keywords $Q_a = w_1, w_2, w_3, ...$ (e.g., "staff,

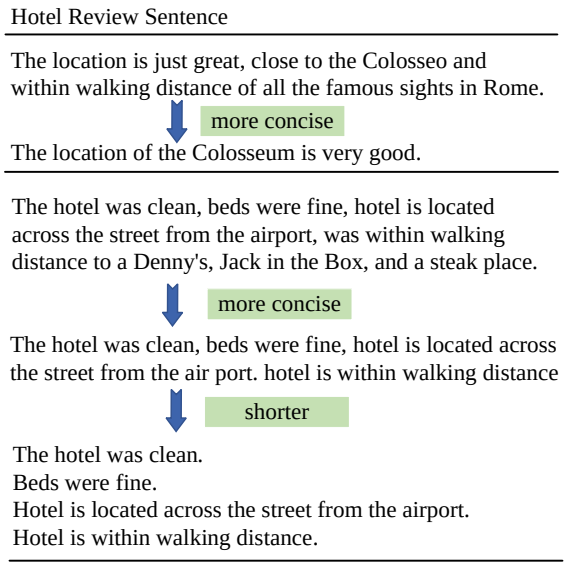


Figure 3: Examples of text simplification. In the first example, the text simplifier retains the core points of the review sentence. In the second example, the review sentence is further divided based on punctuation after being processed by the text simplifier, resulting in a more concise and easier-to-understand summary. For better readability, we capitalize the first letter of each sentence processed by the text simplifier.

friendly")(Angelidis et al., 2021) to filter and select sentences. Through this method, we can obtain a set of sentences $S^(e)_a$ related to a specific aspect. Finally, following the steps previously used to obtain a general summary, we can obtain an opinion summary based on a specific aspect a.

4 Experimental Setup

In this section, we will describe the detailed information of the experimental setup, including datasets, baselines, and other relevant details.

4.1 Datasets

We conduct experiments on two datasets from different domains. SPACE(Angelidis et al., 2021) is a dataset of hotel reviews. It contains manually crafted consensus summaries as well as aspect-based opinion summaries. Amazon(Bražinskas et al., 2020b) is a product review dataset covering topics such as electronics, clothing, health advice, etc. We preprocess these two datasets using the previous preprocessing methods(Angelidis et al., 2021; Bražinskas et al., 2020b). The development and test sets of both datasets contain three human-written general summaries per entity. Space dataset also contains three human-written aspect-specific summaries for six aspects: building, cleanliness, food, location, rooms, and service. The statistics are shown in Table 1.

Data	Ent(Train/Test)	Revs/Ent(Test)
SPACE	11.4k/50	100
Amazon	183k/60	8

Table 1: Dataset statistics for Space and Amazon. The statistics include the number of entities in the training set and the test set Ent (Train / Test), and the average number of comments per entity in the test set Revs / Ent (Test).

4.2 Implementation Details

In the text simplification module, we use the BART model(Lewis et al., 2020) as the backbone model. We fine-tune the BART model on the document-to-document simplification datasets D-Wikipedia and Wiki-Doc(Sun et al., 2021; Kauchak, 2013). We use RoBERTa-base as the sentiment analysis model(Zhuang et al., 2021), and fine-tune it using the SPACE and Amazon datasets. The value of β in the sentiment information penalty term is set to 0.01. In the sentiment classification module, the number of categories K is set to 5. Under the random sampling scheme, the value of λ is 2. In the proportional sampling scheme, the value of φ is 0.1. All experiments are conducted on a GeForce RTX 3090 GPU.

4.3 Baselines

To validate the effectiveness of our model TSSG, we select a series of abstractive and extractive summarization models as our baselines.

We compare TokenCluster with MeanSum(Chu and Liu, 2019), CopyCat(Bražinskas et al., 2020b), DenoiseSum(Amplayo and Lapata, 2020), AceSum(Amplayo et al., 2021a), COOP(Iso et al., 2021), LexRank(Erkan and Radev, 2004), QT(Angelidis et al., 2021), SemAE(Chowdhury et al., 2022), TokenCluster(Li et al., 2023).

5 Results

5.1 Automatic Evaluation

We evaluate the performance of the summarization models using ROUGE-F1 scores(Erkan and Radev, 2004). We present the scores for ROUGE-1 (R1), ROUGE-2 (R2), and ROUGE-L (RL). Table 2 presents the evaluation results of the TSSG model for general opinion summarization on the Amazon dataset. In the evaluation on the Amazon dataset,

Method	R1	R2	RL
Abstractive			
MeanSum	29.20	4.70	18.15
CopyCat	31.97	5.81	20.16
PlanSum	32.87	6.12	19.15
COOP	36.57	7.23	21.24
Extractive			
LexRank	29.78	4.36	13.72
SemAE	30.53	5.02	15.42
QT	32.01	5.23	15.97
HERCULES-ext	30.75	6.04	15.80
TokenCluster	**32.48**	**6.56**	16.78
Ours			
TSSG-weight	31.26	5.70	**16.94**

Table 2: Experimental Results on the Amazon Dataset. The best results for extractive methods are reported in bold, while the best results for abstractive summarization methods are underlined.

Method	R1	R2	RL
Abstractive			
MeanSum	34.95	7.49	19.92
CopyCat	36.66	8.87	20.90
AceSumABS	40.37	11.51	23.23
COOP	-	13.53	26.56
Extractive			
AceSumEXT	35.50	7.82	20.09
LexRank	30.58	5.05	18.72
SemAE	42.75	12.09	24.84
QT	38.66	10.22	21.90
HERCULES-ext	-	13.15	24.43
TokenCluster	**44.05**	12.84	26.61
Ours			
TSSG-random	42.09	12.47	23.34
TSSG-equal	43.18	14.00	26.67
TSSG-weight	43.20	**14.22**	**26.98**

Table 3: Experimental Results on the Space Dataset.The best results are reported in bold.

we report the results for weighted sampling. Table 3 shows the evaluation results of the TSSG model for general opinion summarization on the Space dataset. The results indicate that the TSSG model outperforms other methods in terms of ROUGE-2 and ROUGE-L scores.

Table 4 presents the evaluation results of the TSSG model for aspect-specific opinion summarization on the Space dataset. We report the ROUGE-L score for each aspect and the average ROUGE-1 (R1), ROUGE-2 (R2), and ROUGE-L (RL) scores across all aspects. The TSSG model surpasses the compared methods in the mean R2 and RL metrics.

On the Space dataset, our proposed TSSG model achieves state-of-the-art performance in both general opinion summarization and aspect-based summarization tasks compared to the baselines. We attribute this improvement to the enhanced understanding of sentiment during the inference process and the successful reduction of sentence complexity. However, on the Amazon dataset, we observed a relatively smaller performance improvement brought by the TSSG model. Through analysis, we speculate that this is due to the limited number of reviews per product in the Amazon dataset, averaging only 8, which is much lower than the average number of reviews per entity in the Space dataset (about 100). With fewer sentences expressing conflicting opinions in the original reviews, the performance improvement of the TSSG model is limited.

5.1.1 Sentiment Consistency Evaluation

To evaluate the sentiment consistency of the model's opinion summaries. We calculated the sentiment polarity distribution for both the model-generated summaries and the human-written summaries. We then used KL divergence to measure the similarity between the sentiment distributions of the human-written and model-generated summaries, treating this similarity as the sentiment consistency score. The results are shown in Table 6, where TSSG achieves the best performance in terms of sentiment consistency score.

5.2 Ablation Study

5.2.1 The impact of each module

To validate the effectiveness of the innovations proposed in this paper, we conducted ablation experiments on the general opinion summarization tasks of the Space and Amazon datasets. We individually removed the text simplifier and the sentiment analyzer, or replaced the sentence representation model SemAE with BERT contextual embeddings, and then observed the changes in the evaluation metrics. This approach helps to verify the effectiveness of each component. The results of the ablation experiments are shown in Table 5.

Method	Building	Cleanliness	Food	Location	Rooms	Service	$\overline{R2}$	$\overline{R2}$	$\overline{RL}$
Abstractive									
MeanSum	13.25	19.24	13.01	18.41	17.81	20.40	23.24	3.72	17.02
CopyCat	17.10	15.90	14.53	20.31	17.30	20.05	24.95	4.82	17.53
Extractive									
LexRank	14.73	19.41	18.51	21.13	18.22	24.12	27.76	7.33	19.35
SemAE	18.73	21.52	20.41	23.13	24.31	25.18	29.81	8.93	22.21
QT	16.45	25.12	17.79	23.63	21.26	**26.67**	28.95	8.34	21.77
HERCULES-ext	-	-	-	-	-	-	-	7.93	19.96
TokenCluster	19.81	**25.39**	19.73	25.16	24.86	25.92	**31.17**	9.16	23.57
Ours									
TSSG-weight	**19.24**	24.68	**20.77**	**25.32**	**25.24**	26.35	30.10	**9.69**	**23.65**

Table 4: Evaluation results of aspect-specific summarization on Space dataset. The best results achieved are highlighted in bold.

	Method	R1	R2	RL
Space	TSSG	**43.20**	**14.37**	**26.48**
	w/o simplifier	43.01	13.80	26.60
	w/o sentiment	42.81	12.95	24.91
	w/o semAE	38.22	10.22	21.22
Amazon	TSSG	**31.26**	**5.70**	**16.64**
	w/o simplifier	31.19	5.63	16.31
	w/o sentiment	30.81	5.22	15.90
	w/o semAE	28.32	4.12	14.52

Table 5: Ablation Study Results on Space and Amazon Datasets. We observe that TSSG performs significantly better than the baselines achieving state-of-the-art results.

Method	Sentiment Consistency
Extractive	
LexRank	0.230
SemAE	0.293
QT	0.338
TokenCluster	0.199
TSSG-weight	**0.177**

Table 6: Sentiment Consistency Results on the Space Dataset. TSSG shows a better sentiment consistency score compared to others.

For the general opinion summarization task on the Space dataset, removing the text simplifier results in a decline across all performance metrics, with ROUGE-2 showing the most significant decrease. This indicates that sentence simplification contributes to the quality of the opinion summaries. Removing the guidance from sentiment information results in a noticeable decline in all performance metrics. This demonstrates that sentiment information is crucial for the quality of opinion summaries, as our method reduces the likelihood of non-consensus opinions in the summaries. Without the SemAE model to obtain sentence representations, the performance on both datasets drops significantly. This is because SemAE has been pre-trained on the Space and Amazon datasets and can accurately capture the semantic representations of sentences, which is essential for TSSG. On the Amazon dataset, removing either component leads to a slight decline in all performance metrics, further demonstrating the benefits of the proposed components.

5.2.2 The impact of text simplifyers on length

We also observed the impact of the text simplification module on the length and number of the final opinion summary sentences. We compared the TSSG model variant without the text simplifier to other models, as shown in Table 5. On the Space dataset, the TSSG model's opinion summary results contained 269 sentences, with an average sentence length of 5.9 words. The results indicate that the TSSG model with the text simplifier selects more concise sentences and consequently increases the number of selected sentences. However, this effect was less pronounced on the Amazon dataset. We speculate that this is because the Space dataset contains a significantly larger number of review sentences than the Amazon dataset, with more complex sentences included, leading to better performance.

5.2.3 The Impact of Sentiment Categories

For comparing the impact of different numbers of sentiment polarity classes on opinion summarization, we experimented with varying values of K. We then observed the performance of TSSG on the Space dataset for general opinion summarization.

Method	Space		Amazon	
	Reviews	Avg Len	Reviews	Avg Len
SemAE	202	8.42	292	11.2
TSSG	269	5.9	307	9.6
w/o simplifier	232	7.2	297	9.2

Table 7: Impact of Text Simplification on Opinion Summarization. We report the number of sentences (nums) contained in the opinion summaries and the average length (len) of each sentence in the opinion summaries.

Figure 4 shows the changes in the mean values of ROUGE-1, ROUGE-2, and ROUGE-L scores under different K values. The best performance is achieved when K is set to 5. We speculate that this is because most product review datasets use rating scales within five levels, and setting K to 5 allows the model to more accurately capture the differences between different types of sentences.

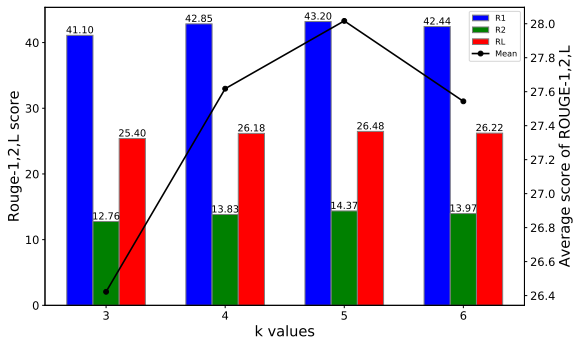


Figure 4: The figure illustrates the performance variation with different values of K. The left y-axis displays the scores for ROUGE-1, ROUGE-2, and ROUGE-L using grouped bar charts. Concurrently, the right y-axis presents the average scores for ROUGE-1, ROUGE-2, and ROUGE-L as a line plot. The best performance is achieved when K is set to 5.

6 Conclusion

In this work, we introduced the TSSG model, a novel two-stage unsupervised opinion summarization method that effectively addresses the challenges of non-consensus opinions and summary conciseness. By leveraging text simplification in the first stage and sentiment guidance in the second stage, our approach successfully extracts representative and sentiment-aligned summary sentences. The experimental results on the Space and Amazon benchmark datasets highlight the strong performance of the TSSG model, demonstrating significant improvements over traditional methods. Despite its success, the model's dependence on input sentiment information can be limiting when dealing with raw data that predominantly contains a single sentiment type. Future research should focus on enhancing sentiment extraction techniques and incorporating aspect-level analysis to further refine the filtering of non-consensus opinions, ultimately improving the overall quality of opinion summaries.

7 Limitations

We propose the TSSG model, a sentence extraction algorithm capable of automatically identifying the polarity of user review sentences, simplifying them, and utilizing this information for extractive summarization. While TSSG effectively addresses the issue of non-consensus opinions in generated summaries and relies on input sentiment information, it may not perform well when the original dataset lacks conflicting opinions or sentiment diversity. This limitation highlights the model's dependency on varied sentiment input for optimal performance.

8 Ethical Consideration

We anticipate that our work will not pose any ethical risks. The datasets we use are all publicly available, and we will not annotate any data ourselves. All the datasets contain user reviews in English.

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